Control and Coordination

The big idea

To survive, an organism must respond to changes around it. In animals, the **nervous system** and **hormones** coordinate these responses. Plants have no nerves, so they use **hormones** and growth movements.

The nervous system

A **neuron** has dendrites (receive information), a cell body and an axon (carries the impulse away). Neurons meet at a **synapse**, where chemicals carry the signal across the gap.

REFLEX ACTION

A quick, automatic response to a stimulus, like pulling your hand off a hot plate. The path is the **reflex arc**: receptor → sensory neuron → **spinal cord** → motor neuron → effector (muscle). The spinal cord responds before the brain even knows.

The human brain

Part	Role
Forebrain (cerebrum)	Thinking, memory, voluntary actions, senses (sight, hearing, smell)
Midbrain	Involuntary actions like reflex movements of the head and eyes

Part	Role
Hindbrain: cerebellum	Balance, posture and precise voluntary movements (riding a cycle)
Hindbrain: medulla	Involuntary actions: blood pressure, salivation, vomiting
Hindbrain: pons	Regulates breathing

The brain is protected by the skull and fluid; the spinal cord by the vertebral column.

Coordination in plants

- **Immediate response**: touch-me-not (Mimosa) leaves fold when touched, by changing the water content of cells.
- **Growth movements (tropisms)**: phototropism (shoots grow towards light), geotropism (roots grow down), hydrotropism (towards water), chemotropism (pollen tube towards the ovule). Tendrils of climbers coil around support (thigmotropism).

PLANT HORMONES

Auxin: made at the shoot tip; on the shaded side it makes cells grow longer, so the shoot bends towards light.
Gibberellins: stem growth. **Cytokinins**: cell division (in fruits and seeds).
Abscisic acid: inhibits growth; causes wilting of leaves.

Hormones in animals

Gland	Hormone	Function
Pituitary	Growth hormone	Body growth (too little → dwarfism, too much → gigantism)

Gland	Hormone	Function
Thyroid	Thyroxin	Metabolism; needs **iodine** (lack → goitre)
Pancreas	Insulin	Lowers blood sugar (lack → diabetes)
Adrenal	Adrenaline	"Fight or flight": faster heartbeat and breathing
Testes / ovaries	Testosterone / oestrogen	Changes at puberty

Feedback mechanism: the timing and amount of hormones are controlled. For example, when blood sugar rises, the pancreas releases more insulin; when it falls, insulin secretion drops.

Quick recap

- Nerve impulses travel neuron to neuron across synapses.
- Reflex arcs pass through the spinal cord for speed.
- Plant movement uses hormones such as auxin.
- Endocrine glands release hormones controlled by feedback.

Practice questions

1. The part of the brain that maintains balance and posture is the
   (a) cerebrum (b) medulla (c) cerebellum (d) pons
2. Iodised salt is recommended to prevent
   (a) diabetes (b) goitre (c) dwarfism (d) anaemia
3. A shoot bends towards light because of
   (a) unequal distribution of auxin (b) gibberellins (c) abscisic acid (d) cytokinins
4. In a reflex arc, the message is processed in the
   (a) cerebrum (b) spinal cord (c) cerebellum (d) receptor
5. Which hormone regulates blood sugar?
   (a) Adrenaline (b) Thyroxin (c) Insulin (d) Growth hormone

Answers: 1. c 2. b 3. a 4. b 5. c